

5. For  $k = N - 1, \dots, 1, 0$ , execute the following RTS smoother equations:

$$\begin{aligned}
 \mathcal{I}_{f,k+1}^- &= \left(P_{f,k+1}^-\right)^{-1} \\
 K_k &= P_{fk}^+ F_k^T \mathcal{I}_{f,k+1}^- \\
 P_k &= P_{fk}^+ - K_k (P_{f,k+1}^- - P_{k+1}) K_k^T \\
 \hat{x}_k &= \hat{x}_{fk}^+ + K_k (\hat{x}_{k+1} - \hat{x}_{f,k+1}^-)
 \end{aligned} \tag{9.138}$$

## 9.5 SUMMARY

In this chapter we derived the optimal smoothing filters. These filters, sometimes called retrodiction filters [Bar01], include the following variants.

- $\hat{x}_{j,k} = E(x_j | y_1, \dots, y_{k-1})$  ( $k \geq j$ ) is the output of the fixed-point smoother. In this filter we find the estimate of the state at the fixed time  $j$  when measurements continue to arrive at the filter at times greater than  $j$ . The time index  $j$  is fixed while  $k$  continues to increase as we obtain more measurements.
- $\hat{x}_{k-N,k} = E(x_{k-N} | y_1, \dots, y_k)$  for a fixed  $N$  is the output of the fixed-lag smoother. In this filter we find the estimate of the state at each time  $k$  while using measurements up to and including time  $(k + N)$ . The time index  $k$  varies while  $N$  remains fixed.
- $\hat{x}_{k,N} = E(x_k | y_1, \dots, y_N)$  for a fixed  $N$  is the output of the fixed-interval smoother. In this filter we find the estimate of the state at each time  $k$  while using measurements up to and including time  $N$ . The time index  $k$  varies while the total number of measurements  $N$  is fixed. The two formulas we derived for this type of smoothing included the forward-backward smoother and the RTS smoother.

Just as steady-state filters can be used for standard filtering, we can also derive steady-state smoothers to save computational effort [Gel74]. An early survey of smoothing algorithms is given in [Med73].

## PROBLEMS

### Written exercises

- 9.1 Prove or disprove the following conjecture: The trace of the inverse of a matrix is equal to the inverse of the trace of the matrix.
- 9.2 Show that  $(A + B)^{-1} = B^{-1}(AB^{-1} + I)^{-1}$ .
- 9.3 Derive Equation (9.83).
- 9.4 Consider a scalar system with  $F = 1$ ,  $H = 1$ , and  $R = 2Q$ .
  - a) What is the steady-state value of the *a priori* estimation-error covariance  $P_k^-$ ?

- b) Suppose that after the Kalman filter has reached steady state, the fixed-point smoother begins to operate. Find a closed-form solution to the covariance of the smoothed estimate  $\Pi_k$  as a function of the time index  $k$ . What is the limiting value of  $\Pi_k$  as  $k \rightarrow \infty$ ?

**9.5** Repeat Problem 9.4 for the case  $R = 12Q$ . What is the percent improvement in the estimation-error covariance due to smoothing? Explain why the percent improvement due to smoothing for this case differs in the way that it does from the results of Problem 9.4.

**9.6** Consider a scalar system with  $F = 1$ ,  $H = 1$ , and  $R = 2Q$ . Suppose that the fixed-lag smoother for this system is in steady state so that  $P_{k+1}^- = P_k^-$ ,  $L_{k+1,i} = L_{k,i}$ ,  $P_{k+1}^{i,i} = P_k^{i,i}$ , and  $P_{k+1}^{0,i} = P_k^{0,i}$ , for  $i = 1, \dots, N+1$ . Find closed-form expressions for  $P_k^-$ ,  $L_{k,i}$ ,  $P_k^{i,i}$ , and  $P_k^{0,i}$  as functions of  $i$ . What is the limit as  $i \rightarrow \infty$  of  $L_{k,i}$ ,  $P_k^{i,i}$ , and  $P_k^{0,i}$ ?

**9.7** Suppose you have a fixed-lag smoother as shown in Equation (9.43) that is in steady state. How do the eigenvalues of the fixed-lag smoother relate to the eigenvalues of the standard Kalman filter? What do you conclude about the stability of the fixed-lag smoother?

**9.8** Solve Equation (9.10) for  $(y_k - H_k \hat{x}_k^-)$  [assuming that  $\rho(L_k) = r$ , where  $r$  is the number of measurements in the system]. Substitute the resulting expression for  $(y_k - H_k \hat{x}_k^-)$  in the fixed-lag smoother equation for  $\hat{x}_{k+1-i,k}$  to show that the smoothed state estimate can be driven by the state estimates without any input from the measurements [And79].

**9.9** Suppose that  $\hat{x}_f$  and  $\hat{x}_b$  are unbiased estimates of  $x$ , and  $\hat{x} = K_f \hat{x}_f + K_b \hat{x}_b$ . Show that if  $\hat{x}$  is an unbiased estimate of  $x$ , then we must have  $K_f + K_b = I$ .

**9.10** Consider a scalar system with  $F = 1$ ,  $H = 1$ , and  $R = 2Q$ . Use the forward-backward smoother of Section 9.4.1 to find the steady-state value of the covariance of the smoothed state estimate.

**9.11** Consider a scalar system with  $F = 1$ ,  $H = 1$ , and  $R = 2Q$ . Use the RTS smoother of Section 9.4.2 to find the steady-state value of the covariance of the smoothed state estimate.

**9.12** Consider a scalar system with  $F = 1$ ,  $H = 1$ , and  $R = 2Q$ . Suppose that the forward filter has reached steady state. Use the RTS smoother of Section 9.4.2 to find the covariance of the smoothed state estimate for  $k = N, N-1, N-2, N-3$ , and  $N-4$ . At what point does the covariance of the smoothed state estimate get within 1% of its steady-state value?

**9.13** Repeat Problem 9.12 for  $R = 12Q$ . How do you intuitively explain the quicker convergence of  $P_k$  to steady state?

**9.14** Use the RTS smoother equations to show that constant states are not smoothable. That is, if  $F = I$  and  $Q = 0$ , then  $P_k = P_{fN}^+$  for all  $k$ .

### Computer exercises

**9.15** Consider the second-order system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\zeta\omega \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w(t)$$

where  $\omega = 6$  rad/s is the natural frequency of the system, and  $\zeta = 0.16$  is the damping ratio. The input  $w(t)$  is continuous-time white noise with a variance of 0.01. Measurements of the first state are taken every 0.5 s:

$$y(t_k) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t_k) + v(t_k)$$

where  $v(t_k)$  is discrete-time white noise with a variance of  $10^{-4}$ . The initial state, estimate, and covariance are

$$\begin{aligned} x(0) &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ \hat{x}(0) &= x(0) \\ P(0) &= \begin{bmatrix} 10^{-5} & 0 \\ 0 & 10^{-2} \end{bmatrix} \end{aligned}$$

- a) Discretize the system equation.
  - b) Implement the discrete-time Kalman filter and the RTS smoother for 10 s (20 time steps). Plot the variance of the estimation error of the first state for the forward filter and for the RTS smoother on a single plot. Do the same for the second state. Why is the second state more smoothable than the first state?
- 9.16** Repeat Problem 9.15 with the continuous-time process noise  $w(t)$  having a variance of 1. How does this change the smoothability of the states?
- 9.17** Design a fixed-interval smoother for the system described in Problem 5.11 to estimate the state at each time on the basis of measurements at all 10 time steps.
- a) Plot the *a posteriori* covariance of the forward state estimate and the covariance of the smoothed state estimate as a function of time for both states.
  - b) What are the percent improvements in the estimation-error variances due to smoothing for the two states at the initial time? Why is there so much more improvement for one state than for the other state?
  - c) Simulate the system and smoother a hundred times or so, each simulation with a different noise history. On the basis of your simulations, derive a numerical estimate of the smoother estimation-error variances of the two states at the initial time. How do your numerical variances compare with the theoretical variances obtained in part (b)?